



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

FPGA BASED SYSTEM DESIGN [ELE-4411]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) List the four classes of commercially available FPGAs.

(2)

A)

B)

Analyze the Verilog code given below to construct the synthesized circuit.
For addition and comparison operations, use adder and comparator blocks.

```
module addborcb(x, a, b, c, d);
```

```
input wire [15:0] a, b, c;
```

```
input wire d;
```

```
output reg [15:0] x;
```

```
reg [15:0] t;
```

```
always @ ( a or b or c or d ) begin
```

```
if ( d ) t = b; else t = c;
```

```
if ( a < 8 ) t = t + 12;
```

```
x = a + t;
```

```
end
```

```
endmodule
```

(4)

C)

Implement the Boolean function of sum of full adder using ACT-1 module.
ACT-1 module is given by Fig. 1C.

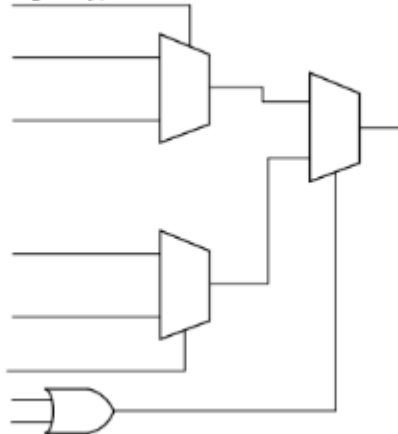


Fig. 1C

(4)

- 2) Illustrate an application of FPGA in embedded system design.

(2)

A)

- B) Design a 5-bit Linear Feedback Shift Register (LFSR) that generates a maximum-length sequence. Modify the circuit so that the all-zero state (00000) is included in the sequence without causing the sequence to lock up. Provide the updated feedback logic and determine the complete state sequence, including the all-zero state. (4)
- C) Determine the number of LUTs required to implement a digital system for an elevator safety mechanism that prevents the elevator from moving unless the following conditions are met: (4)
1. The elevator door is completely closed.
 2. The weight sensor detects that the weight is within the safe limit.
 3. The floor selection button is pressed.
 4. Additionally, a self-test mechanism must ensure that when the elevator is powered ON, the system momentarily allows movement for 5 seconds, even if the above conditions are not met, to verify system functionality.

Give the truth table implemented in each LUT. Name the input output pins selected on a FPGA layout.

- 3) List the limitations of FPGA. (2)

- A)
- B) Write Verilog test bench code for half adder. Display the message "error" if half adder outputs are not matching with expected functionality. Assume the Verilog code for the design under test is available with necessary input and output. (4)
- C) For the circuit given by Fig. 3C, determine the set of test vectors that detect only all stuck-at-0 faults in the circuit. Make sure to use the minimum number of vectors. (4)

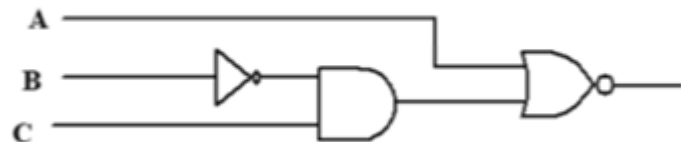


Fig. 3C

- 4) List one advantage and one disadvantage of coarse-grained FPGA architecture compared to fine-grained FPGA architecture. (2)
- A)
- B) A digital system is required to control an automatic sliding door with the following conditions. An input 'p' to the digital system indicates whether a sensor detects a person in front of the door. An input 'q' indicates whether the door should be manually held open regardless of whether a person is detected. An input 'r' indicates whether the door should be forced to stay closed. r=1 means the door should stay closed. Assume a Verilog module is available describing the behavior of the digital system. Write the Verilog self-checking test bench code to check the functionality of the Verilog module. Check for all the combinations of the inputs. (4)
- C) Design a digital system to multiply two 3-bit unsigned numbers using a Spartan 2E FPGA. (4)
- Tasks:
- i. Implement the system using only LUTs (Look-Up Tables).
 - ii. Implement the system using LUTs and dedicated carry logic or multipliers, if available.
- For each implementation:

- Calculate the number of LUTs required.
- Determine the number of logic levels (depth of the logic).
- Estimate the number of Configurable Logic Blocks (CLBs) used.

Assume each CLB in Spartan 2E contains 4 LUTs and supports fast carry logic.

5) Consider the logic circuit given in Fig. 5A. Choose the best LUT to implement the logic circuit.

Available LUTs:

- A)
- 4-input, 1-output LUT with delay of 1.5 ns
 - 6-input, 1-output LUT with delay of 3 ns

Assume all signals are stable and ignore fan-out effects.

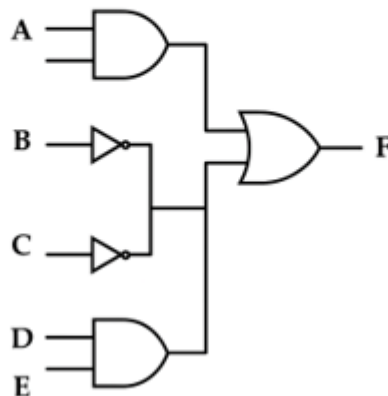


Fig. 5A

(2)

B) Design a digital logic system for a motorcycle dashboard that turns ON a helmet warning indicator. The indicator should light up under the following conditions:

- The ignition key is turned ON.
- The rider is seated (detected via a seat sensor).
- The helmet is not detected (via an RFID helmet sensor).

Additionally, to confirm that the indicator is functional, it should automatically turn ON for the first 5 seconds after the ignition is turned ON, regardless of the other inputs. (3)

Tasks:

- Design the logic circuit using basic combinational logic gates.
- Show the truth table for the system.
- Draw the stick diagram representing the implementation layout of the logic.

C) The impulse response of a linear phase FIR filter is $h(n)=[1 \ 2]$. Develop FPGA based architecture for two-bit parallel distributed arithmetic FIR filter. Explain the developed architecture. Determine response of developed filter for the input $x(n)=[4 \ 5]$. Provide an explanation for the partial outputs corresponding to each bit shift operation of the shift register. (5)

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